SADAB SAZZAD ALIF

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PROFILE

Software Engineer with a background in Computer Engineering and currently pursuing an MSc at Cardiff University, specialising in full-stack development, cloud computing, and scalable systems. Experienced in developing web and application-based solutions across multiple programming languages, with a strong focus on performance, scalability, and clean architecture. Skilled in modern frameworks and DevOps practices, delivering robust and efficient software in fast-paced environments.

EDUCATION

Master's in Software Engineering with Professional Placement Year
Cardiff University – 2024 to 2027

- Relevant Modules: Software Architecture, Artificial Intelligence, Data Science, Cybersecurity, Agile Software Development, Web Applications, DevOps, Data Mining, Principles in Programming

Bachelor's in Computer Engineering
Asia Pacific University – 2019 to 2023

- CGPA: 3.3 / 4.0
- Relevant Modules: Data Structures & Algorithms, Software Engineering Principles, Machine Learning, Cloud Computing, Cybersecurity, Computer Networks, Artificial Intelligence, Database Management Systems
- **Final Year Project:** Optimized contact tracing application using a hybrid genetic algorithm

A-Levels

- Mathematics – A, Physics – B, Chemistry – B

O-Levels

- Mathematics B – B, Physics – A, Chemistry – A, Biology – A, Further Mathematics – B, Bangla – A, English – A

PROJECTS

AI-Based Contact Tracing System

- Designed and implemented a hybrid genetic algorithm to optimise tracing accuracy and efficiency
- Improved processing speed and reduced computational cost compared to traditional approaches
- Built using Python with a focus on algorithm optimisation and scalability

Full-Stack Web Application (SaaS Platform)

- Developed a scalable web platform using React, Node.js, and MongoDB
- Implemented authentication (JWT) and RESTful APIs for secure data handling
- Deployed using Docker and AWS, ensuring high availability and performance

Cloud-Based DevOps Pipeline

- Designed CI/CD pipelines using GitHub Actions and Jenkins
- Automated deployment workflows, reducing release time by over 50%
- Integrated Docker and Terraform for infrastructure automation

Java-Based Library Management System

- Developed an object-oriented system using Java and Spring Boot for managing books, users, and transactions
- Implemented CRUD operations and RESTful APIs for efficient data handling
- Applied design patterns (e.g., Singleton, MVC) to improve maintainability and scalability

Python Data Analysis & Automation Tool

- Built a data processing tool using Python (Pandas, NumPy) to analyse large datasets and generate insights
- Automated repetitive data tasks, reducing manual processing time significantly
- Visualised results using Matplotlib to support decision-making

C Programming System Utilities (Low-Level Projects)

- Developed multiple system-level programs in C, including file handling and memory management utilities
- Implemented data structures (linked lists, stacks, queues) from scratch to strengthen core programming fundamentals
- Optimised code for performance and memory efficiency

PROFESSIONAL EXPERIENCE

Software Engineer (2026 – Present)

Welsh Government

- Contributing to the development and maintenance of software applications within the Welsh Government's Data & Geography team
- Supporting the development of digital solutions and data-driven tools to improve public services
- Applying software engineering principles and modern development technologies to build reliable and maintainable applications
- Collaborating with developers, colleagues, and stakeholders to understand requirements and deliver effective technical solutions
- Participating in the software development lifecycle, including requirements analysis, implementation, testing, and deployment
- Exploring geospatial technologies, data visualisation, and innovative approaches to enhance digital services and support evidence-based decision-making

Software Engineer (2023)

Sugar Book

- Developed scalable software solutions aligned with business objectives
- Contributed to system architecture design to enhance performance and reliability

- Collaborated with cross-functional teams to deliver high-quality features
- Led and mentored junior engineers, improving code quality and team productivity
- Implemented microservices architecture, increasing system scalability by 30%

Software Engineer (2020 – 2022)

Zap Lab Studios

- Optimized database queries, reducing load times by 40% and improving user experience
- Designed and implemented CI/CD pipelines, accelerating release cycles by 50%
- Developed and maintained robust APIs, enhancing system integration and interoperability
- Automated infrastructure provisioning using IaC tools such as Terraform and CloudFormation
- Strengthened application security through authentication and authorization mechanisms
- Supported cloud adoption strategies, reducing infrastructure costs by 25%
- Produced detailed technical documentation to improve team collaboration and onboarding

TECHNICAL SKILLS

Programming Languages:

- Proficient: Python, JavaScript, Java, SQL
- Working Knowledge: TypeScript, C++

Software Development:

- Full-Stack Development: React.js, Node.js, Express.js, MongoDB, Next.js
- Backend Frameworks: Spring Boot, Django, Flask, FastAPI
- Frontend: HTML5, CSS3, Tailwind CSS, Bootstrap, Redux

Cloud & DevOps:

- AWS (EC2, S3, Lambda)
- Docker, Kubernetes
- CI/CD: GitHub Actions, Jenkins
- Infrastructure as Code: Terraform, CloudFormation

Database Management:

- SQL: PostgreSQL, MySQL
- NoSQL: MongoDB, Redis

Testing & Quality Assurance:

- Unit Testing: PyTest, JUnit
- Integration Testing: Postman, Selenium
- Code Quality Tools: ESLint, SonarQube

Development Practices:

- Agile (Scrum), CI/CD, REST API Development

Version Control & Collaboration:

- Git (GitHub), Code Reviews, Jira

Monitoring & Security:

- Monitoring: Grafana, Prometheus
- Security: JWT, OAuth 2.0, basic OWASP principles

Soft Skills:

- Problem-solving, Communication, Teamwork, Adaptability

ACHIEVEMENTS

Best Innovation in Hackathon – Cardiff University, 2024

Recognized for developing an AI-driven tool for optimizing cloud resource usage.

PROFESSIONAL MEMBERSHIPS

Student Member – Software Engineering & Technology Community

Actively developing expertise in software engineering, application development, geospatial technologies, and data-driven solutions. Keeping abreast of emerging technologies, modern development practices, and industry innovations to support continuous professional growth.

EXTRACURRICULAR ACTIVITIES & INTERESTS

- Participated in hackathons and coding competitions, enhancing problem-solving and teamwork skills
- Write tech blogs on programming best practices and emerging technologies
- Conduct software development and basic cybersecurity workshops
- Mentor junior students in coding and software development concepts
- Passionate about football (playing and following the sport)
- Interest in digital art and design, with a focus on creativity and visual storytelling

REFERENCES

Dr Fernando Loizides

Reader (Associate Professor) and Director of the Data Science Academy

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